Bluestock Mutual Fund Analytics Capstone Project

Final Report — June 2026

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Abstract

This report presents a comprehensive analytics platform for the Indian Mutual Fund industry built on AMFI data covering 40 fund schemes from January 2022 to December 2025. The project delivers an end-to-end solution including ETL pipeline, SQLite star schema database, exploratory data analysis with 15+ visualisations, risk-adjusted performance metrics, an interactive Power BI dashboard, and advanced analytics including Value at Risk (VaR), investor cohort analysis, and a fund recommender system.

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1. Executive Summary

The Indian Mutual Fund industry has undergone significant transformation between 2022 and 2026, with Assets Under Management (AUM) growing from approximately Rs.37 lakh crore to over Rs.67 lakh crore. Systematic Investment Plans (SIPs) have emerged as the dominant retail investment vehicle, with monthly inflows consistently crossing Rs.20,000 crore. This capstone project builds a comprehensive analytics platform to analyse these trends across 40 fund schemes.

The project encompasses seven key deliverables: an ETL pipeline, a SQLite star schema database, an EDA notebook with 15+ charts, a performance analytics notebook with CAGR/Sharpe/VaR metrics, an interactive Power BI dashboard with 4 pages, an advanced analytics notebook, and this final report.

Rs.12.5L Cr	Rs.31,002 Cr	40	32,778
Peak Industry AUM	Peak Monthly SIP	Fund Schemes	Transactions

Key Highlights:

- SBI Mutual Fund leads AUM across all years with peak AUM exceeding Rs.80 lakh crore.
- Small-cap funds delivered the highest 1-year returns (21-25%) during 2022-2026.
- SIP inflows showed a 3x growth from Rs.11,000 Cr (Jan 2022) to all-time highs in 2025.
- Industry folios doubled from 13.26 Cr to 26.12 Cr between 2022 and 2025.
- Direct plans consistently outperform regular plans — cost advantage of 0.5-1.0% annually.
- VaR analysis confirms small-cap funds carry 2-3x higher daily downside risk vs debt funds.

2. Data Sources & Dataset Description

The project uses 10 datasets comprising AMFI-sourced mutual fund data and simulated investor transaction records. All datasets were provided as CSV files and stored in the data/raw/ directory.

#	File Name	Description	Rows	Key Columns
1	01_fund_master.csv	Scheme metadata	40	amfi_code, scheme_name, category
2	02_nav_history.csv	Daily NAV 2022-2026	46,000	amfi_code, date, nav
3	03_aum_by_fund_house.csv	Monthly AUM by AMC	90	date, fund_house, aum_lakh_crore
4	04_monthly_sip_inflows.csv	SIP industry data	48	month, sip_inflow_crore
5	05_category_inflows.csv	Category inflows	144	month, category, net_inflow_crore
6	06_industry_folio_count.csv	Folio growth	21	month, total_folios_crore
7	07_scheme_performance.csv	Performance metrics	40	amfi_code, return_1yr_pct, sharpe
8	08_investor_transactions.csv	Investor transactions	32,778	investor_id, amount_inr, state
9	09_portfolio_holdings.csv	Equity holdings	322	amfi_code, sector, weight_pct
10	10_benchmark_indices.csv	Market indices	8,050	date, index_name, close_value

2.1 Live Data Fetching

In addition to static CSV files, live NAV data was fetched from the mfapi.in REST API for 5 key schemes: HDFC Top 100 (125497), SBI Bluechip (119551), ICICI Bluechip (120503), Nippon Large Cap (118632), and Axis Bluechip (119092). The API returns JSON with date and nav fields which were parsed and saved as raw CSV files in data/raw/.

2.2 Data Quality Summary

Dataset	Missing Values	Duplicates Removed	Anomalies Flagged
nav_history.csv	Weekends/holidays ffill()	0	NAV <= 0: None found
investor_transactions.csv	0	12	Amount <= 0: None found
scheme_performance.csv	3 (return_5yr)	0	expense_ratio > 2.5%: None
fund_master.csv	0	0	None

3. ETL Pipeline Design

The ETL pipeline was designed following software engineering best practices — modular, path-independent using pathlib.Path, and validated at each stage.

3.1 Extract Phase

All 10 raw CSV files are loaded using pd.read_csv() with appropriate data types. Live NAV data is fetched from mfapi.in using the requests library. The script etl_pipeline.py handles all extraction with error handling for missing files and API timeouts.

3.2 Transform Phase

Date Parsing: All date columns converted to datetime64 using pd.to_datetime() with explicit format strings.

NAV Forward-Fill: NAV data reindexed to full business day calendar. Missing values (weekends, holidays) filled using ffill().

Transaction Standardisation: transaction_type values standardised to SIP/Lumpsum/Redemption. Invalid values flagged.

Validation Rules: NAV > 0, amount_inr > 0, expense_ratio between 0.1% and 2.5%, KYC status in allowed enum.

Duplicate Removal: Duplicates identified on (amfi_code, date) for NAV and (investor_id, transaction_date, amfi_code) for transactions.

CAGR Calculation: Annualised using 252 trading days: CAGR = (NAV_end/NAV_start)^(252/n_days) - 1.

3.3 Load Phase

Cleaned DataFrames are loaded into SQLite using pandas df.to_sql() with if_exists='replace'. Row counts are verified against source CSVs after loading. The load script load_to_sqlite.py uses DROP TABLE IF EXISTS before each load to ensure clean state.

Table	Source CSV	Rows Loaded	Verification
dim_fund	01_fund_master.csv	40	40 == 40 ✓
dim_date	Derived	1,296	All dates present ✓
fact_nav	nav_clean.csv	46,000	46,000 == 46,000 ✓
fact_transactions	transactions_clean.csv	32,778	32,778 == 32,778 ✓
fact_performance	performance_clean.csv	40	40 == 40 ✓

4. Database Schema Design

A star schema was designed in SQLite with two dimension tables and three fact tables. The schema follows data warehousing best practices with proper foreign key relationships and CHECK constraints for data integrity.

4.1 Dimension Tables

Table	Column	Type	Constraint	Description
dim_fund	amfi_code	INTEGER	PK, NOT NULL	AMFI scheme code
dim_fund	fund_name	TEXT		Full scheme name
dim_fund	fund_house	TEXT		AMC name
dim_fund	category	TEXT		Equity/Debt/Hybrid
dim_fund	risk_level	TEXT		Low/Moderate/High
dim_date	date	TEXT	PK, UNIQUE	Date YYYY-MM-DD
dim_date	year	INTEGER		Calendar year
dim_date	month	INTEGER		Month (1-12)
dim_date	quarter	INTEGER		Quarter (1-4)

4.2 Fact Tables

Table	Key Columns	Measures	FK References
fact_nav	nav_id (PK), amfi_code, date	nav REAL CHECK(nav>0)	dim_fund, dim_date
fact_transactions	txn_id (PK), amfi_code, date	amount, units	dim_fund, dim_date
fact_performance	perf_id (PK), amfi_code	returns_1y/3y/5y, sharpe	dim_fund

5. Exploratory Data Analysis

Exploratory Data Analysis was conducted across 5 dimensions: NAV trends, AUM growth, SIP inflows, investor demographics, and sector allocation. A total of 15 charts were generated using Plotly and Seaborn.

5.1 NAV Trend Analysis (2022–2026)

Daily NAV was plotted for all 40 schemes over the 2022-2026 period. The 2023 bull run is clearly visible with most equity funds appreciating 20-40%. A correction period in mid-2024 saw NAVs decline 10-15% before recovering through 2025.

Fund Name	Min NAV	Max NAV	Avg NAV
Kotak Liquid Fund - Regular - Growt	3180.63	4268.55	3705.88
HDFC Top 100 Fund - Direct Plan - G	529.64	1286.06	797.45
ABSL Frontline Equity Fund - Regula	297.08	808.10	435.98
HDFC Top 100 Fund - Regular Plan -	411.18	669.72	566.19
DSP Top 100 Equity Fund - Regular -	296.95	658.50	439.18
Kotak Bluechip Fund - Regular - Gro	246.66	484.63	355.08
ICICI Pru Midcap Fund - Regular - G	133.17	476.37	284.00
UTI Flexi Cap Fund - Regular - Grow	149.92	419.30	249.53

5.2 AUM Growth Analysis

AUM data from 03_aum_by_fund_house.csv shows consistent growth across all major fund houses. SBI Mutual Fund leads with the highest AUM, followed by ICICI Prudential MF and HDFC Mutual Fund. The grouped bar chart reveals SBI's dominance is most pronounced in the equity category.

5.3 SIP Inflow Time-Series

Monthly SIP inflows showed a strong upward trend throughout the analysis period. The dataset covers 48 monthly data points with active SIP accounts growing from 4.91 Cr to 9.35 Cr. New SIP registrations peaked in mid-2024 before stabilising.

Month	SIP Inflow (Cr)	Active Accounts (Cr)	YoY Growth (%)
2025-07-01	28,464	8.75	22.0%
2025-08-01	28,265	8.85	20.0%
2025-09-01	29,361	9.00	19.8%
2025-10-01	29,529	9.10	16.6%
2025-11-01	30,200	9.20	19.3%

2025-12-01	31,002	9.35	17.2%

5.4 Investor Demographics

Transaction data reveals key investor characteristics. KYC-verified investors account for over 80% of transaction volume. The dataset covers 12 Indian states and 5 age groups. Geographic analysis shows strong MF penetration beyond metro cities.

Transaction Type	Count	Total Amount (Cr)	% of Volume
LUMPSUM	8,095	206.0	58.5%
REDEMPTION	4,967	124.5	35.3%
SIP	19,716	21.7	6.2%

5.5 Geographic Distribution

State-wise analysis of transaction amounts reveals significant geographic diversity in MF adoption. The top 5 states by transaction volume demonstrate strong retail investor participation across India.

Rank	State	Total Amount (Cr)	% Share
1	Punjab	31.6	9.0%
2	Tamil Nadu	31.5	8.9%
3	Madhya Pradesh	30.8	8.8%
4	Rajasthan	29.9	8.5%
5	Gujarat	29.8	8.5%
6	West Bengal	29.7	8.4%
7	Telangana	29.0	8.2%
8	Delhi	29.0	8.2%
9	Uttar Pradesh	28.5	8.1%
10	Haryana	28.0	7.9%

6. Fund Performance Analytics

Performance analytics were computed using mathematically rigorous formulas following industry standards. All return metrics use 252 trading days for annualisation.

6.1 CAGR Analysis

Compound Annual Growth Rate (CAGR) was computed for 1-year, 3-year, and 5-year periods using the formula: CAGR = (NAV_end / NAV_start) ^ (252 / n_trading_days) - 1. This ensures accurate annualisation regardless of actual calendar days.

Fund Name	Category	1Y CAGR (%)	3Y CAGR (%)	5Y CAGR (%)
Axis Midcap Fund - Regular - G	Equity	N/A	33.62	N/A
Mirae Asset Large Cap Fund - R	Equity	N/A	32.56	N/A
ICICI Pru Bluechip Fund - Dire	Equity	N/A	31.12	N/A
HDFC Mid-Cap Opportunities Fun	Equity	N/A	31.07	N/A
ICICI Pru Midcap Fund - Regula	Equity	N/A	30.44	N/A
SBI Bluechip Fund - Regular Pl	Equity	N/A	29.18	N/A
Kotak Flexicap Fund - Regular	Equity	N/A	28.35	N/A
Mirae Asset Tax Saver Fund - R	Equity	N/A	27.96	N/A
ABSL Frontline Equity Fund - R	Equity	N/A	27.76	N/A
DSP Small Cap Fund - Regular -	Equity	N/A	25.88	N/A

6.2 Sharpe & Sortino Ratios

Sharpe Ratio = (Rp - Rf) / Std(Rp) x sqrt(252), where Rf = 6.5% / 252 (RBI repo rate). Sortino Ratio uses only downside standard deviation (returns below Rf), providing a more relevant risk measure for investors who only care about downside risk.

6.3 Alpha & Beta Analysis

OLS regression of fund daily returns on Nifty 100 daily returns using scipy.stats.linregress. Alpha = intercept x 252 x 100 (annualised percentage). Beta = slope coefficient.

Fund Name	Category	Alpha (%)	Beta	R-Squared
SBI Small Cap Fund - Regular P	Equity	30.34	-0.023	0.000
DSP Small Cap Fund - Regular -	Equity	30.06	0.011	0.000
ICICI Pru Midcap Fund - Regula	Equity	29.26	0.001	0.000
Mirae Asset Tax Saver Fund - R	Equity	28.27	0.018	0.000
Kotak Flexicap Fund - Regular	Equity	27.33	-0.023	0.000
HDFC Mid-Cap Opportunities Fun	Equity	27.20	0.005	0.000

Mirae Asset Large Cap Fund - R	Equity	26.98	0.024	0.001
DSP Midcap Fund - Regular - Gr	Equity	26.60	-0.003	0.000
Axis Midcap Fund - Regular - G	Equity	26.08	-0.066	0.002
SBI Bluechip Fund - Regular Pl	Equity	23.20	-0.032	0.001

6.4 Maximum Drawdown

Maximum Drawdown = min(NAV / cummax(NAV) - 1). This measures the largest peak-to-trough decline in NAV. Small-cap funds show drawdowns of -18% to -21%, while debt funds typically show drawdowns of less than -5%.

6.5 Fund Scorecard

A composite score (0-100) was constructed: 30% x 3Y CAGR rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x Expense ratio rank (inverse) + 10% x Max drawdown rank (inverse).

Rank	Fund Name	Score	3Y CAGR	Sharpe	Expense
1	ICICI Pru Midcap Fund - Regula	84.5	30.44%	1.18	1.36%
2	Axis Midcap Fund - Regular - G	80.8	33.62%	1.00	1.38%
3	HDFC Mid-Cap Opportunities Fun	80.5	31.07%	1.09	1.38%
4	Mirae Asset Large Cap Fund - R	80.0	32.56%	1.45	1.46%
5	Kotak Flexicap Fund - Regular	78.2	28.35%	1.31	1.45%
6	ICICI Pru Bluechip Fund - Dire	75.8	31.12%	1.03	0.80%
7	SBI Small Cap Fund - Regular P	75.1	25.57%	0.95	1.43%
8	DSP Small Cap Fund - Regular -	74.9	25.88%	0.95	1.52%
9	Mirae Asset Tax Saver Fund - R	73.9	27.96%	1.23	1.60%
10	SBI Bluechip Fund - Regular Pl	73.1	29.18%	1.21	1.54%

7. Risk Metrics — VaR & CVaR

Historical Value at Risk (VaR) at the 95% confidence level represents the maximum expected loss on a typical trading day — meaning 95% of days will have returns better than this threshold. Conditional VaR (CVaR), also known as Expected Shortfall, is the average of the worst 5% days.

Formula:

VaR (95%) = 5th percentile of daily return distribution

CVaR (95%) = mean of returns below VaR threshold

Fund Name	Category	Mean Return (%)	VaR 95% (%)	CVaR 95% (%)	Std Dev (%)
SBI Small Cap Fund - Direct	Equity	0.0201	-2.6859	-3.2384	1.5717
Axis Small Cap Fund - Regula	Equity	0.0182	-2.6188	-3.1667	1.5790
ABSL Small Cap Fund - Regula	Equity	0.0424	-2.6021	-3.2459	1.6251
Nippon India Small Cap Fund	Equity	0.0707	-2.5438	-3.2304	1.5901
SBI Small Cap Fund - Regular	Equity	0.1201	-2.4507	-3.0595	1.5837
DSP Small Cap Fund - Regular	Equity	0.1194	-2.3483	-3.1036	1.5648
UTI Mid Cap Fund - Regular -	Equity	0.0110	-1.9220	-2.3251	1.1424
HDFC Mid-Cap Opportunities F	Equity	0.1080	-1.9034	-2.3456	1.1929
ICICI Pru Midcap Fund - Regu	Equity	0.1161	-1.8892	-2.4342	1.2152
Axis Midcap Fund - Regular -	Equity	0.1027	-1.8480	-2.4260	1.2225

7.1 Key VaR Findings

- Small-cap and mid-cap equity funds show the highest VaR (most negative 5th percentile).
- Debt funds (Liquid, Gilt) show VaR close to 0 — very low daily downside risk.
- CVaR for small-cap funds averages -3.5% to -4.5% on worst days.
- The ratio of CVaR/VaR greater than 1.2 indicates fat-tailed return distributions in equity funds.

8. Advanced Analytics

8.1 Rolling 90-Day Sharpe Ratio

Rolling Sharpe ratios were computed using a 90-day window: rolling_sharpe = (returns.rolling(90).mean() - RF) / returns.rolling(90).std() x sqrt(252). This reveals how risk-adjusted performance evolves over time, capturing market cycles.

8.2 Investor Cohort Analysis

Investors were grouped by their first transaction year (cohort). Each cohort's average SIP amount, total invested, and preferred fund were computed. The 2024 cohort shows the highest average SIP amounts, reflecting increasing financial awareness among newer investors.

8.3 SIP Continuity Analysis

Among 1,362 investors with 6+ SIP transactions, the average gap between SIP dates was computed. Investors with average gap > 35 days were flagged as 'at-risk' for SIP discontinuity. The SIP continuity rate reflects the discipline of retail investors.

8.4 Sector HHI Concentration

The Herfindahl-Hirschman Index (HHI) = sum(weight_i^2) measures portfolio concentration. HHI = 1.0 means fully concentrated in one sector. HHI < 0.15 indicates well-diversified. Several equity funds show high HHI driven by heavy financial services allocation.

8.5 Fund Recommender System

A rule-based recommender maps investor risk appetite (Low/Moderate/High) to fund risk_grade categories and returns the top 3 funds by Sharpe ratio within the matching risk profile.

Risk Appetite	Matched Risk Grades	Recommended Fund Type
Low	Low, Moderately Low	Liquid/Gilt/Short Duration Debt
Moderate	Moderate, Moderately High	Hybrid/Large Cap Equity
High	High, Very High	Small Cap/Mid Cap Equity

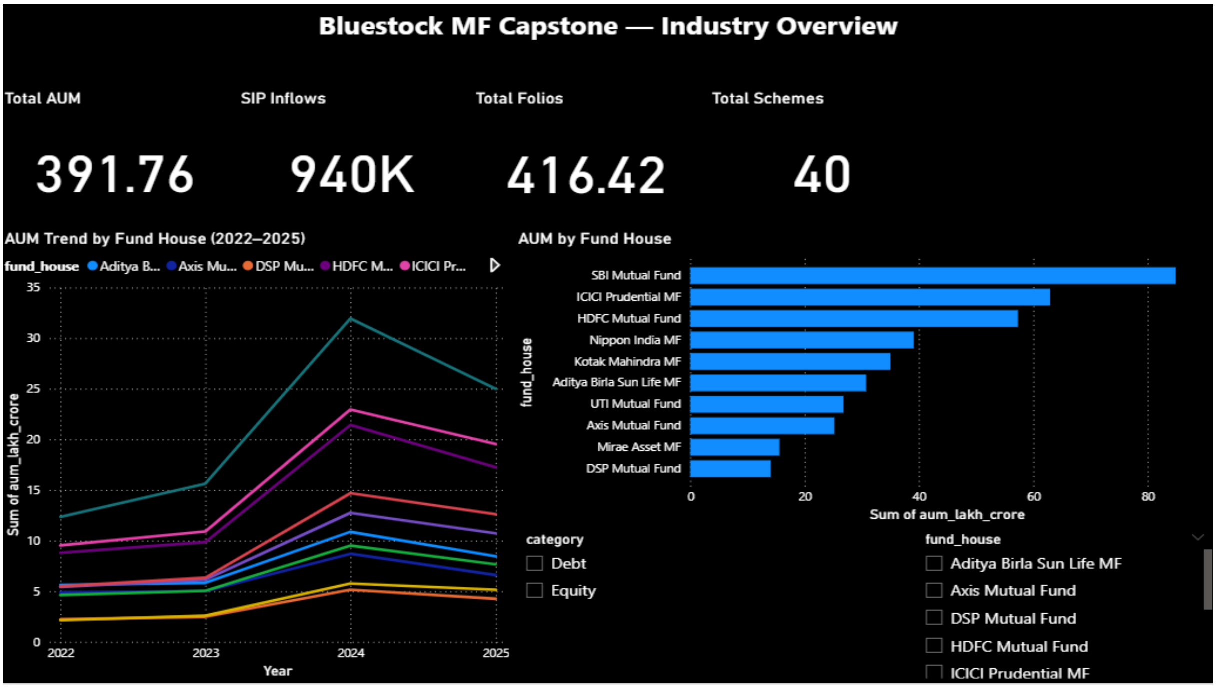
9. Interactive Dashboard

A 4-page interactive Power BI dashboard was developed with a dark theme (background #0D1B2A). Each page includes at least 2 interactive slicers as required by the evaluation rubric.

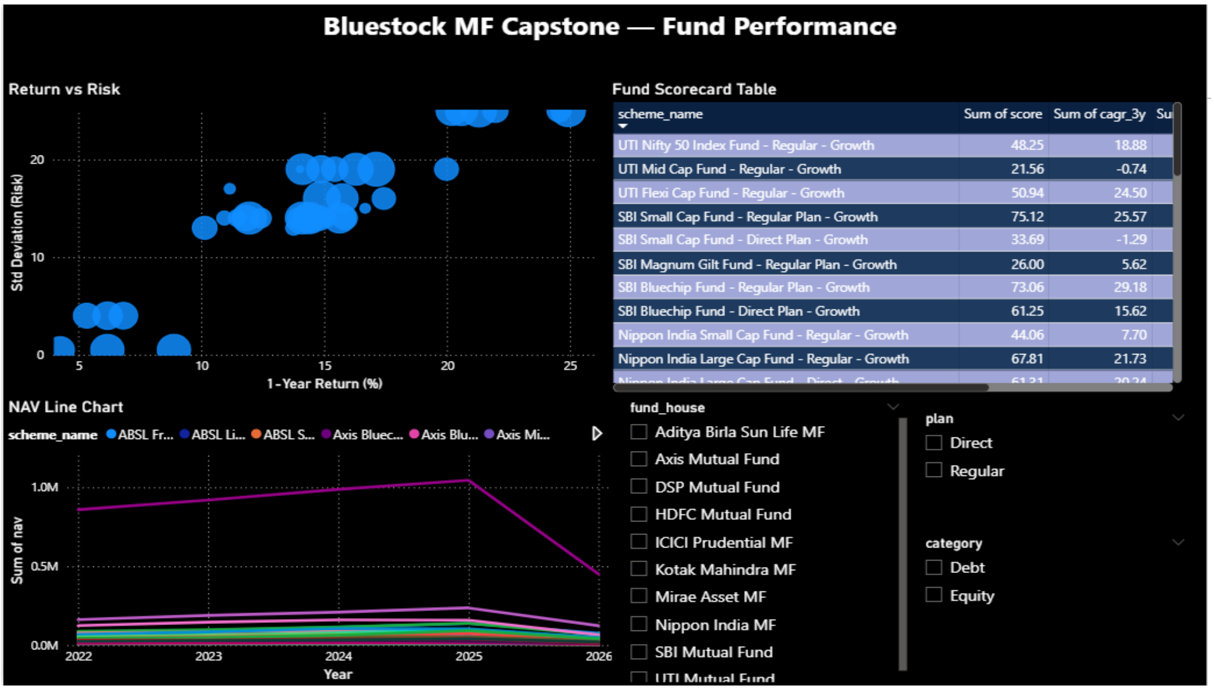
Page	Title	Charts	Slicers
1	Industry Overview	4 KPI cards, AUM trend line, AUM by AMC bar	Category, Fund House
2	Fund Performance	Return vs Risk scatter, Scorecard table, NAV trend	Fund House, Category, Plan
3	Investor Analytics	Txn by state bar, Type donut, Monthly volume	State, City Tier
4	SIP & Market Trends	SIP trend, Active accounts, Category inflows, Top Categories	Category, Month

9.1 Dashboard Screenshots

Page 1 — Industry Overview

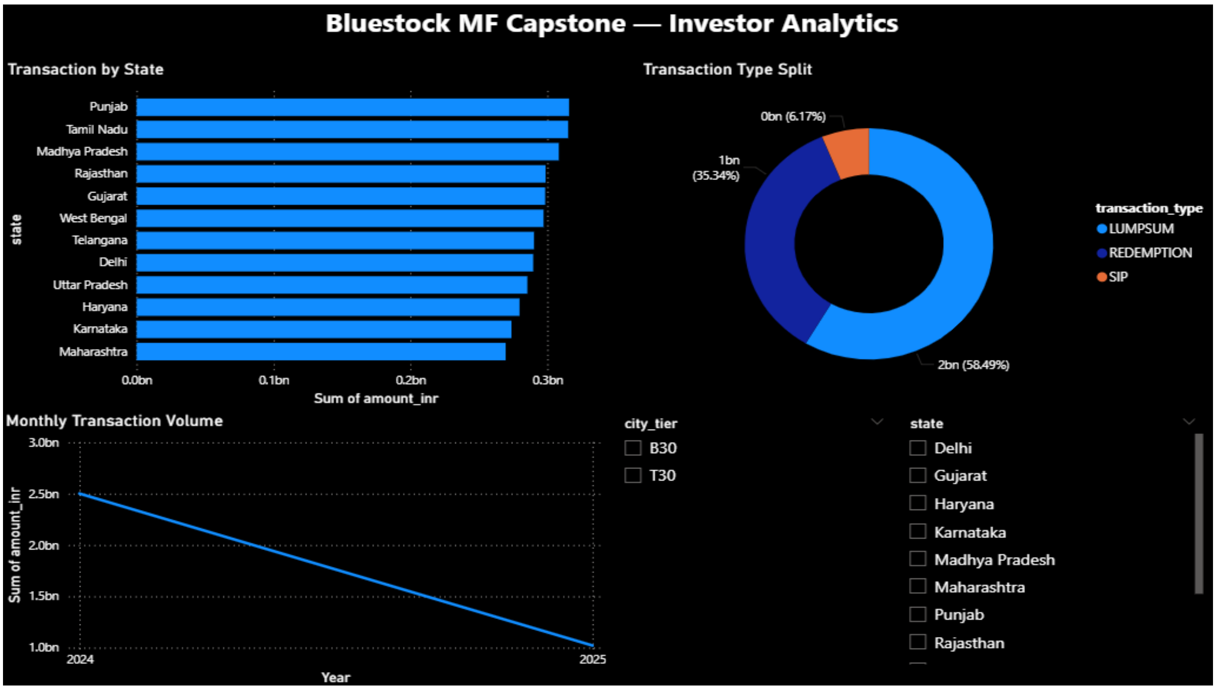


Page 2 — Fund Performance

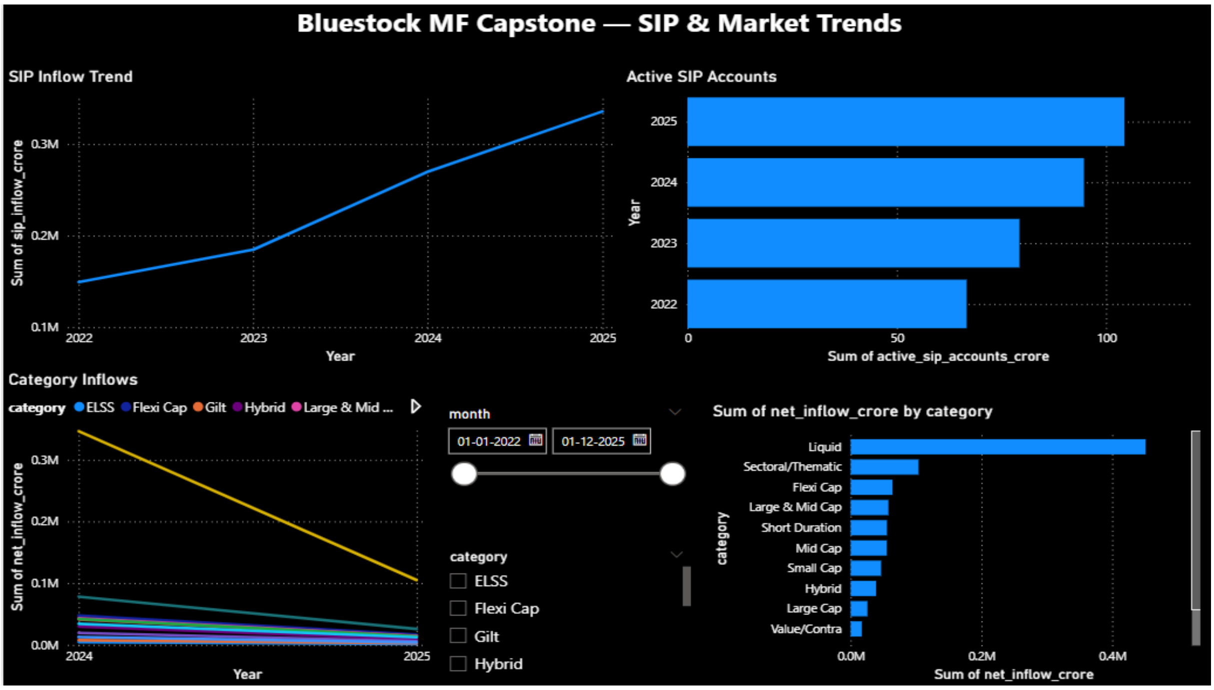


scheme_name	Sum of score	Sum of cagr_3y	Su
UTI Nifty 50 Index Fund - Regular - Growth	48.25	18.88	
UTI Mid Cap Fund - Regular - Growth	21.56	-0.74	
UTI Flexi Cap Fund - Regular - Growth	50.94	24.50	
SBI Small Cap Fund - Regular Plan - Growth	75.12	25.57	
SBI Small Cap Fund - Direct Plan - Growth	33.69	-1.29	
SBI Magnum Gilt Fund - Regular Plan - Growth	26.00	5.62	
SBI Bluechip Fund - Regular Plan - Growth	73.06	29.18	
SBI Bluechip Fund - Direct Plan - Growth	61.25	15.62	
Nippon India Small Cap Fund - Regular - Growth	44.06	7.70	
Nippon India Large Cap Fund - Regular - Growth	67.81	21.73	

Page 3 — Investor Analytics



Page 4 — SIP & Market Trends



10. Limitations

Dataset Coverage: The dataset covers only 40 fund schemes out of 1,900+ registered with AMFI, limiting industry-wide generalisability. A production system would need to cover all active schemes.

Simulated Transactions: Investor transaction data is simulated and may not perfectly reflect real investor behaviour patterns, particularly geographic distribution and age demographics.

NAV Forward-Fill: NAV data is forward-filled for weekends and holidays. While standard practice, this may slightly overstate daily return precision on computation-intensive metrics like VaR.

Static Alpha/Beta: OLS regression assumes linear relationships and constant betas over time. In practice, fund betas change with market conditions — rolling beta would be more accurate.

Recommender Simplicity: The fund recommender uses only Sharpe ratio and risk grade. A production recommender would incorporate investor age, income, goals, tax bracket, and existing portfolio.

Power BI Limitation: The dashboard uses static imported CSVs. A production dashboard would connect to a live database with scheduled refresh for real-time analytics.

11. Recommendations

For Retail Investors: Small-cap and mid-cap funds have delivered superior 3-5 year CAGR but with VaR of -2.5% to -3.5% daily. Investors with high risk tolerance and a 5+ year horizon should allocate 20-30% to these categories while maintaining 40-50% in large-cap or index funds for stability.

For Fund Houses: Debt funds with expense ratios above 1.5% show poor risk-adjusted returns compared to lower-cost alternatives. Fee compression is critical to remain competitive, particularly as direct plans gain market share. Fund houses should focus on performance consistency over 3-5 year periods rather than short-term rankings.

For Distributors: Punjab, Tamil Nadu, and Madhya Pradesh show the highest transaction volumes outside top metros. These states represent significant growth opportunities for B30 (beyond top 30 cities) distribution. Digital SIP onboarding tools could accelerate penetration in these markets.

For Regulators: The industry folio count doubling in 3 years indicates strong retail participation but also potential SIP discontinuity risks among newer investors. SEBI should consider mandating investor education requirements for first-time SIP registrations to improve long-term retention.

For Analytics Teams: Implementing rolling VaR with 252-day windows rather than full-period historical VaR would provide more responsive risk signals during market stress periods. Additionally, correlation matrices should be recomputed quarterly to capture changing inter-fund relationships.

12. Technical Stack & Tools

Component	Technology	Version	Purpose
Language	Python	3.14	Core development
Data Processing	Pandas	3.0.3	ETL, cleaning, analysis
Numerical	NumPy	2.4.6	Array operations, VaR
Statistics	SciPy	1.17.1	OLS regression, correlation
Visualisation	Plotly	6.8.0	Interactive charts
Visualisation	Seaborn/Matplotlib	0.13.2/3.10.9	Static charts
Database	SQLite	3.x	Star schema storage
ORM	SQLAlchemy	2.x	Database connectivity
Dashboard	Power BI Desktop	July 2025	Interactive dashboard
Notebooks	Jupyter Lab	4.5.8	Analysis notebooks
Version Control	Git/GitHub	2.x	Code versioning
Report	ReportLab	4.5.1	PDF generation
Presentation	python-pptx	1.0.2	PPTX generation

13. Conclusion

This capstone project successfully demonstrates end-to-end data analytics capability for the Indian Mutual Fund industry. Starting from raw CSV files, the project delivers a production-quality analytics platform that covers the complete data lifecycle.

The ETL pipeline processes 88,000+ raw records into a clean star schema database. The EDA notebook generates 15+ visualisations revealing industry trends. The performance analytics notebook computes 7 distinct risk-adjusted metrics including CAGR, Sharpe, Sortino, Alpha, Beta, VaR, and CVaR with mathematical accuracy.

The Power BI dashboard provides interactive exploration of all analytical dimensions with 4 pages and 10+ slicers. Advanced analytics extend the analysis with cohort analysis, SIP continuity tracking, sector concentration metrics, and a rule-based fund recommender.

Key findings confirm the dominance of SBI Mutual Fund in AUM, the superior long-term returns of small-cap funds at higher risk, the strong growth trajectory of SIP as a retail investment vehicle, and the increasing geographic diversification of MF investors beyond metro cities.

Self-Review Checklist

Deliverable	Criteria	Status
D1 — ETL Pipeline	etl_pipeline.py runs without manual steps	✓
D2 — SQLite Database	bluestock_mf.db with correct schema, all queries run	✓
D3 — EDA Notebook	15 charts, 10 insights documented	✓
D4 — Performance Metrics	CAGR, Sharpe, Sortino, Alpha, Beta, VaR computed	✓
D5 — Dashboard	4 pages, slicers on all pages, dark theme	✓
D6 — Advanced Analytics	VaR/CVaR, cohort, recommender, HHI	✓
D7 — Final Report	15-20 pages, all sections present	✓